

Seminar Report

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Harmonic Analysis on Euclidean Spaces

June 24, 2025

This report, mainly based on Walter Rudin's article *Trigonometric Series with Gaps* [1], discusses a remarkable result based on some works of Zygmund in *Trigonometrical Series* [2]. The result states that under the condition of a sufficiently “thin” spectrum (such as that of a Sidon set) strong integrability properties hold for functions with frequencies in that set. Specifically, integrability bounds hold for L^p norms and even extend to exponential integrability.

More precisely, if $E \subset \mathbb{Z}$ is a Sidon set, then for every E -function f whose spectrum is contained in E , we have $f \in L^p(\mathbb{T})$ for all $p \geq 1$, and moreover,

$$\int_{\mathbb{T}} \exp(|f(x)|^2) dx < \infty.$$

Historical Context Antoni Zygmund (1900-1992) was a foundational figure in modern harmonic analysis. One of his early successes, which brought him to the attention of mathematicians around the world, was *Trigonometrical Series*, published in Warsaw in 1935. The book quickly became a reference in the field. Jean-Pierre Kahane reportedly referred to Zygmund's *Trigonometrical Series* as a “Bible” for harmonic analysts.

Influenced in the 1930s by G.H. Hardy and J.E. Littlewood during his time in Cambridge, Zygmund would in turn influence and collaborate with his students, most notably Alberto Calderón, Elias M. Stein, and many others.

Walter Rudin (1921-2010) was also influenced by Zygmund and his intellectual successors. He is best known for his widely used textbooks in mathematical analysis: *Functional Analysis*, *Principles of Mathematical Analysis*, nicknamed “Baby Rudin” and *Real and Complex Analysis*, often referred to as “Big Rudin”.

Both Zygmund and Rudin were European-born and immigrated to the United States in response to political instability and anti-Semitic persecution in Europe. Zygmund left Poland, and Rudin, born in Vienna, fled Austria after the Nazi annexation. He spent time in France and England before settling in the United States.

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1 Preliminaries

Let f be a Lebesgue integrable function on the torus $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Its Fourier series has the form:

$$\sum_{n \in \mathbb{Z}} \hat{f}(n) e^{2\pi i n x}$$

where the Fourier coefficient are given by :

$$\hat{f}(n) = \int_{\mathbb{T}} f(x) e^{-2\pi i n x} dx$$

We define the $L^p(\mathbb{T})$ norm in the usual way:

$$\|f\|_p = \left(\int_{\mathbb{T}} |f(x)|^p dx \right)^{1/p}.$$

When $p = \infty$, this corresponds to the supremum norm:

$$\|f\|_{\infty} = \sup_{x \in \mathbb{T}} |f(x)|.$$

We now introduce the concept of functions supported on a set $E \subset \mathbb{Z}$.

Definition 1.1. Let $E \subset \mathbb{Z}$. A function f is called an E -function if and only if $f \in L^1(\mathbb{T})$ and $\hat{f}(n) = 0$ for all $n \notin E$. We denote by $C_E(\mathbb{T})$ the space of all continuous E -functions.

Remark 1.1. In the definition above, the fact that $f \in L^1(\mathbb{T})$ guarantees the existence of its Fourier coefficients. The set E can then be interpreted as the *frequency support* of the function f .

Definition 1.2. A trigonometric polynomial whose Fourier frequencies belong to a set $E \subset \mathbb{Z}$ is called an E -polynomial.

We now introduce the notion of *Sidon*¹ sets, which are special subsets of the integers with useful boundedness properties for Fourier coefficients.

Definition 1.3. A set $E \subset \mathbb{Z}$ is called a *Sidon set* if there exists a constant $S > 0$ such that for every E -polynomial f , one has

$$\sum_{n \in E} |\hat{f}(n)| \leq S \|f\|_{\infty}.$$

The smallest such constant S is called the *Sidon constant* of the set E .

¹Simon Sidon (1892-1941) was a Hungarian mathematician known for his contributions to harmonic analysis. In the early 1930s, he introduced the concept that would later become known as a Sidon set, although the term itself was only formalized after his death, notably by researchers such as Kahane and Rudin in the 1950s and 60s.

2 Characterization Theorem

We now state and prove a fundamental theorem characterizing Sidon sets.

Theorem 2.1 (Characterization of Sidon Sets). *Let $E \subset \mathbb{Z}$. The following statements are equivalent:*

- (i) E is a Sidon set.
- (ii) Every bounded E -function has an absolutely convergent Fourier series.
- (iii) Every continuous E -function has an absolutely convergent Fourier series.
- (iv) For every bounded function $b : E \rightarrow \mathbb{C}$, there exists a measure μ on $\mathbb{T}$ such that:

$$\hat{\mu}(n) = b(n), \quad \forall n \in E;$$

- (v) If $b(n) \rightarrow 0$ as $|n| \rightarrow \infty$, then there exists $f \in L^1(\mathbb{T})$ such that $\hat{f}(n) = b(n)$ for all $n \in E$.

Proof. We prove each implication step by step.

(i) $\Rightarrow$ (ii): Assume that $E \subset \mathbb{Z}$ is a Sidon set with Sidon constant S , and let f be a bounded E -function. Consider the N -th Fejér means $\sigma_N f$ of f , defined by

$$\sigma_N f(x) = \frac{1}{N+1} \sum_{n=0}^N S_n f(x),$$

where $S_n f$ denotes the n -th partial sum of the Fourier series of f . Explicitly,

$$\sigma_N f(x) = \sum_{|n| \leq N} \left(1 - \frac{|n|}{N+1}\right) \hat{f}(n) e^{2\pi i n x}.$$

This is a trigonometric polynomial with frequencies in E , i.e., an E -polynomial. Moreover, we can write

$$\sigma_N f = f * F_N,$$

where F_N is the Fejér kernel of order N . Since convolution with the Fejér kernel preserves uniform boundedness, we have

$$\|\sigma_N f\|_\infty \leq \|f\|_\infty.$$

Therefore,

$$\frac{1}{2} \sum_{|n| \leq N/2} |\hat{f}(n)| \leq \sum_{|n| \leq N} \left(1 - \frac{|n|}{N+1}\right) |\hat{f}(n)| \leq S \|f\|_\infty,$$

where the right-hand inequality follows from the definition of Sidon set. Letting $N \rightarrow \infty$, we conclude that the Fourier series of f converges absolutely.

(ii) $\Rightarrow$ (iii): Since every continuous function on $\mathbb{T}$ is bounded, the implication follows directly.

(iii) $\Rightarrow$ (i): Assume that every continuous E -function has an absolutely convergent Fourier series. Define the map:

$$\begin{aligned} \mathcal{F}_E &: C_E(\mathbb{T}, \|\cdot\|_\infty) \longrightarrow \ell^1(E, \|\cdot\|_1) \\ f &\mapsto \widehat{f}|_E \end{aligned}$$

This map is linear, and by the Closed Graph Theorem, it is therefore continuous. Consequently, there exists a constant $S > 0$ such that for every $f \in C_E(\mathbb{T})$,

$$\sum_{n \in E} |\widehat{f}(n)| \leq S \|f\|_\infty.$$

In particular, this holds for all E -polynomials, proving that E is a Sidon set.

Remark 2.1. In the definition of a Sidon set, replacing E -polynomials with $C_E(\mathbb{T})$ does not affect the definition.

Thus (i), (ii), (iii) are equivalent.

(i) $\Rightarrow$ (iv): The key tools here are the Hahn–Banach theorem and the Riesz representation theorem. We recall both:

Theorem 2.2 (Hahn-Banach). *Let X be a complex normed vector space, $M \subset X$ a linear subspace, and $\mathcal{L} : M \rightarrow \mathbb{C}$ a bounded linear map. Then there exists $\tilde{\mathcal{L}} : X \rightarrow \mathbb{C}$, linear and bounded, such that $\tilde{\mathcal{L}}|_M = \mathcal{L}$ and $\|\tilde{\mathcal{L}}\| = \|\mathcal{L}\|$.*

Theorem 2.3 (Riesz-Markov-Kakutani Representation). *Let X be a compact Hausdorff space. Then for every bounded linear functional $\mathcal{L}$ on $C(X)$, there exists a bounded complex Radon measure ν such that $\mathcal{L}(f) = \int_X f d\nu$ for all $f \in C(X)$.*

Let E be a Sidon set with Sidon constant S . Let $b : E \rightarrow \mathbb{C}$ be bounded, and define a linear functional $\mathcal{B}$ on $C_E(\mathbb{T})$ by:

$$\mathcal{B}(f) = \sum_{n \in E} \widehat{f}(n) b(n).$$

This functional is bounded on $C_E(\mathbb{T})$ with norm at most $\|b\|_\infty S$, since

$$|\mathcal{B}(f)| \leq \|b\|_\infty \sum_{n \in E} |\widehat{f}(n)| \leq S \|b\|_\infty \|f\|_\infty.$$

Remark 2.2. Although $f \in C_E(\mathbb{T})$ is not necessarily an E -polynomial, the Sidon inequality still holds. See Remark 2.1 for details.

By the Hahn–Banach theorem, $\mathcal{B}$ extends to a bounded linear functional on $C(\mathbb{T})$ with the same norm. By the Riesz theorem, there exists a complex Radon measure ν on $\mathbb{T}$, of total variation at most $S\|b\|_\infty$, such that

$$\mathcal{B}(f) = \int_{\mathbb{T}} f(x) d\nu, \quad \forall f \in C_E(\mathbb{T}).$$

Applying this to $f_n(x) = e^{2\pi i n x}$ for all $n \in E$, and considering the measure $\mu(x) = \nu(-x)$, we obtain:

$$\hat{\mu}(n) = \int_{\mathbb{T}} e^{2\pi i n x} d\nu(x) = b(n), \quad \forall n \in E.$$

(iv) $\Rightarrow$ (v): Let $b : E \rightarrow \mathbb{C}$ satisfy $b(n) \rightarrow 0$ as $|n| \rightarrow \infty$. Choose an even, positive, convex sequence $(c(n))_{n \in \mathbb{Z}}$ such that $c(n) \rightarrow 0$ and $\frac{b(n)}{c(n)}$ is bounded on E . This is possible by ensuring $c(n)$ decays more slowly than $b(n)$. By (iv), there exists a measure μ on $\mathbb{T}$ such that

$$\hat{\mu}(n) = \frac{b(n)}{c(n)}, \quad \forall n \in E.$$

Now we use the following lemma:

Lemma 2.1. *Let $(a_n)_{n \in \mathbb{Z}}$ be an even, convex, positive sequence with $a_n \rightarrow 0$. Then there exists $g \in L^1(\mathbb{T})$ such that $\hat{g}(n) = a_n$ for all $n \in \mathbb{Z}$.*

Let $g \in L^1(\mathbb{T})$ be the function given by the lemma applied to $(c(n))_{n \in \mathbb{Z}}$. Define $f = g * \mu \in L^1(\mathbb{T})$. Then:

$$\hat{f}(n) = \hat{g}(n)\hat{\mu}(n) = c(n) \cdot \frac{b(n)}{c(n)} = b(n), \quad \forall n \in E.$$

Thus, $f \in L^1(\mathbb{T})$ and has the desired Fourier coefficients.

(v) $\Rightarrow$ (iii): Let (v) hold and assume $f \in C_E(\mathbb{T})$. We aim to show that for every sequence $(a_n)_{n \in E}$ with $a_n \rightarrow 0$ as $|n| \rightarrow \infty$, the series

$$\sum_{n \in E} |\hat{f}(n)a_n|$$

is convergent.

Lemma 2.2. *If for all sequences $(a_n) \rightarrow 0$, the series $\sum |\hat{f}(n)a_n|$ converges, then $\sum |\hat{f}(n)|$ converges.*

Let $(a_n)_{n \in E}$ be such that $a_n \rightarrow 0$ as $|n| \rightarrow \infty$. By (v), there exists $g \in L^1(\mathbb{T})$ such that $\hat{g}(n) = a_n$ for all $n \in E$. Since $f \in C_E(\mathbb{T})$, the convolution $f * g \in C_E(\mathbb{T})$. By Fejér's theorem, the Fejér means of $f * g$ converge pointwise to $f * g(x)$ for every $x \in \mathbb{T}$, and in particular:

$$\sum_{|n| \leq N} \left(1 - \frac{|n|}{N+1}\right) \hat{f}(n)\hat{g}(n) \rightarrow f * g(0).$$

Assume $\hat{g}(n) = a_n$ cancels the phase of $\hat{f}(n)$, so $\hat{f}(n)a_n \geq 0$. Then:

$$\frac{1}{2} \sum_{|n| \leq N/2} |\hat{f}(n)a_n| \leq \sum_{|n| \leq N} \left(1 - \frac{|n|}{N+1}\right) \hat{f}(n)\hat{g}(n).$$

As the right-hand side converges, so does $\sum_{n \in E} |\hat{f}(n)a_n|$, which completes the proof. $\square$

It remains to prove the two lemma:

Proof of Lemma 2.1. Let $(a_n)_{n \in \mathbb{Z}}$ be an even, convex, positive sequence such that $a_n \rightarrow 0$. By the convexity of the sequence, for all $n \geq 1$, we have:

$$(a_{n-1} - a_n) - (a_n - a_{n+1}) = a_{n-1} + a_{n+1} - 2a_n \geq 0,$$

So the sequence $(a_n - a_{n+1})_{n \geq 0}$ is monotonically decreasing in n . Moreover, observe that:

$$\sum_{n=0}^{\infty} (a_n - a_{n+1}) = a_0.$$

From the two previous facts, we deduce that:

$$n(a_n - a_{n+1}) \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

otherwise the series $\sum_{n=0}^{\infty} (a_n - a_{n+1})$ would diverge. Now, define:

$$f = \sum_{n=1}^{\infty} n(a_{n-1} + a_{n+1} - 2a_n)F_{n-1},$$

where F_n denotes the Fejér kernel of order n . First, note that:

$$\sum_{n=1}^N n(a_{n-1} + a_{n+1} - 2a_n) = a_0 - a_N - N(a_N - a_{N+1}) \rightarrow a_0 \quad \text{as } N \rightarrow \infty,$$

and since $\|F_{n-1}\|_1 = 1$, the series defining f converges in $L^1(\mathbb{T})$. Moreover, the Fourier coefficients of f are given by:

$$\widehat{f}(k) = \sum_{n=|k|+1}^{\infty} n(a_{n-1} + a_{n+1} - 2a_n) \left(1 - \frac{|k|}{n}\right) = a_{|k|}.$$

Remark 2.3. The last equality can be proved by induction on $m \geq 1$, showing that:

$$\sum_{n=|k|+1}^{|k|+m} n(a_{n-1} + a_{n+1} - 2a_n) \left(1 - \frac{|k|}{n}\right) = a_{|k|} + m a_{|k|+m+1} - (m+1)a_{|k|+m}.$$

□

Proof of Lemma 2.2. Suppose that $\sum_{n \in \mathbb{Z}} |\widehat{f}(n)| = \infty$. We construct a sequence $(a_n) \rightarrow 0$ such that $\sum_n |\widehat{f}(n)a_n| = \infty$. Let $\{I_k\}_{k \in \mathbb{N}}$ be a partition of $\mathbb{Z}$ into finite, symmetric, disjoint subsets such that

$$\sum_{n \in I_k} |\widehat{f}(n)| \geq 1 \quad \text{for all } k.$$

Define

$$a_n = \begin{cases} \frac{1}{k+1} & \text{if } n \in I_k, \\ 0 & \text{otherwise.} \end{cases}$$

Then $a_n \rightarrow 0$, but

$$\sum_{n \in \mathbb{Z}} |\hat{f}(n)a_n| = \sum_{k \geq 0} \frac{1}{k+1} \sum_{n \in I_k} |\hat{f}(n)| \geq \sum_{k \geq 1} \frac{1}{k} = \infty.$$

This proves the contrapositive. Hence, if $a_n \rightarrow 0$, then $\sum |\hat{f}(n)a_n| < \infty$.

□

3 An example: Hadamard Sets

We have just proven a characterization theorem for Sidon sets. Before going further, we may want to examine some examples of what a Sidon set can be. For this, we will use one of the characterizations of Sidon sets. Let us introduce a category of sets named after Hadamard².

Definition 3.1. A set of positive integers $H := \{h_n\}$ is called Hadamard lacunary (or simply lacunary) if there exists $r > 1$ such that

$$\frac{h_{n+1}}{h_n} \geq r \quad \forall n \geq 0.$$

Remark 3.1. A trigonometric series is called lacunary if all the frequencies appearing in it are of the form $\pm h_n$.

Example 3.1. A classical example of a Hadamard set is the sequence of powers of an integer $r > 1$:

$$H = \{r^n : n \geq 0\}.$$

Theorem 3.1. *Every Hadamard set is a Sidon set.*

Remark 3.2. Not all Sidon sets are of Hadamard type, as shown in [1, Section 2.5, p. 210].

The general idea is that the sparsity of the exponents in the series imposes a certain uniformity in the behavior of the functions built from them.

Proof. Let f be an H -polynomial:

$$f(x) = \sum_{n=0}^N c_n e^{2\pi i h_n x}.$$

By the triangle inequality, it suffices to prove the result for $\Re f$ and $\Im f$. These functions have Fourier series of the form:

$$\sum_{|n| \leq N} \tilde{c}_n e^{2\pi i h_n x}, \quad \text{with } h_{-n} = -h_n.$$

We define the Riesz product:

$$P_N(x) = \prod_{n=1}^N (1 + \cos(2\pi h_n x + \phi_n)),$$

where the phases $\phi_n \in \mathbb{T}$ are arbitrary.

Case $r > 3$: In this case, every integer $k \in \mathbb{Z}$ admits a unique representation of the form:

$$k = \sum \epsilon_n h_n, \quad \epsilon_n \in \{-1, 0, 1\}.$$

²Jacques Hadamard (1865-1963) was a French mathematician known for his work in complex analysis, number theory, and functional analysis. These sets are named after him due to his theorem on Hadamard gaps in trigonometric series.

This uniqueness ensures that in the expansion of $P_N(x)$ given below, the second sum does not introduce any of the frequencies h_n already present in the first sum:

$$P_N(x) = 1 + \sum_{n=1}^N \cos(2\pi h_n x + \phi_n) + \sum A_k \cos(2\pi k x + \phi_k),$$

where the second sum runs over those integers k expressible as above with at least two nonzero ϵ_n , and the coefficients A_k are constants. Thus, for each $k \in \mathbb{Z}$:

$$\widehat{P}_N(k) = \delta_{k=0} + \sum_{n=1}^N \frac{1}{2} (e^{i\phi_n} \delta_{k=h_n} + e^{-i\phi_n} \delta_{k=-h_n}) + \sum A_k (e^{i\phi_k} \delta_{k=n} + e^{-i\phi_n} \delta_{k=-n}).$$

In particular,

$$\|P_N\|_1 = \widehat{P}_N(0) = 1, \quad \widehat{P}_N(h_n) = \frac{1}{2} e^{i\phi_n}, \quad \widehat{P}_N(-h_n) = \frac{1}{2} e^{-i\phi_n}.$$

We choose the phases ϕ_n so that

$$\overline{\tilde{c}_n} e^{i\phi_n} = |\tilde{c}_n|.$$

Since P_N , $\Re f$, and $\Im f$ are continuous (and hence in $L^2(\mathbb{T})$), we apply Parseval's identity:

$$\frac{1}{2} \sum |\tilde{c}_n| = \frac{1}{2} \sum \overline{\tilde{c}_n} e^{i\phi_n} \leq \int_{\mathbb{T}} P_N(x) \cdot \overline{\Re f(x)} dx \leq \|\Re f\|_{\infty}.$$

The same estimate holds for $\Im f$, hence we conclude:

$$\sum |c_k| \leq 4\|f\|_{\infty}.$$

Case $1 < r < 3$: In this case, there may exist nontrivial relations of the form:

$$h_k = \sum \epsilon_n h_n.$$

To reduce to the previous case, choose $M \in \mathbb{N}$ large enough such that:

$$r^M > 3, \quad 1 - \frac{1}{r^M - 1} > \frac{1}{r}, \quad 1 + \frac{1}{r^M - 1} < r.$$

Define the subsequences:

$$h_n^{(m)} := h_{m+nM}, \quad \phi_n^{(m)} := \phi_{m+nM}.$$

Then,

$$h_{n+1}^{(m)} > r^M h_n^{(m)}.$$

We construct the corresponding Riesz product:

$$P_N^{(m)}(x) = \prod_{n=1}^N (1 + \cos(2\pi h_n^{(m)} x + \phi_n^{(m)})),$$

with $\phi_n^{(m)} \in \mathbb{T}$ as before. Its expansion takes the same form:

$$P_N^{(m)}(x) = 1 + \sum_{n=1}^N \cos(2\pi h_n^{(m)} x + \phi_n^{(m)}) + \sum A_k \cos(2\pi k x + \phi_k),$$

where $k = \sum \epsilon_n h_n^{(m)}$ and the A_k are constants. Now observe that any $k = \sum \epsilon_n h_n^{(m)}$ is at a distance less than $\frac{h_{n_0}^{(m)}}{r^M - 1}$ from some $h_{n_0}^{(m)}$. Indeed, if $h_{n_0}^{(m)}$ is the dominant term, then by lacunarity:

$$|k - h_{n_0}^{(m)}| \leq \sum_{n < n_0} h_n^{(m)} < h_{n_0}^{(m)} \sum_{l=1}^{\infty} \frac{1}{r^{Ml}} = \frac{h_{n_0}^{(m)}}{r^M - 1}.$$

The only potential issue is if a linear combination of the $h_n^{(m)}$ coincides exactly with some h_k outside the subsequence. We now prove this never happens, that is, for $k \not\equiv m \pmod{M}$, the frequencies $\pm h_k$ do not appear in the Riesz product. Indeed, if $|h_k - h_{n_0}^{(m)}| < \frac{h_{n_0}^{(m)}}{r^M - 1}$ for some n_0 , then:

$$\frac{1}{r} < 1 - \frac{1}{r^M - 1} < \frac{h_k}{h_{n_0}^{(m)}} < 1 + \frac{1}{r^M - 1} < r.$$

Moreover, by lacunarity, if $k < m + n_0 M$, then

$$\frac{h_k}{h_{n_0}^{(m)}} < \frac{1}{r},$$

and if $k > m + n_0 M$, then

$$\frac{h_k}{h_{n_0}^{(m)}} > r,$$

In both cases, this contradicts the above inequality, hence such h_k cannot appear. Thus, the only frequencies h_k that appear in the Riesz product satisfy:

$$k \equiv m \pmod{M}.$$

Consequently, by repeating the argument from the $r > 3$ case, we deduce that

$$\sum_n |c_{m+nM}| \leq 4\|f\|_{\infty}.$$

Summing over $m = 1, \dots, M$, the result follows. $\square$

Remark 3.3. A more conceptual understanding of the role of the Riesz product arises from duality: according to characterization (iv) of Sidon sets, the Riesz product provides an explicit construction of the interpolating measures involved.

4 Norm Inequalities for Sidon Sets

The characterization theorem established earlier in Section 2 is not only conceptually important but also technically useful. In particular, the construction of interpolating measures is a key tool in proving norm inequalities. The inequality we are about to prove directly leads us to the desired result of exponential integration.

Theorem 4.1. *Let $E \subset \mathbb{Z}$ be a Sidon set with Sidon constant S . Then, for any E -polynomial f , if $2 < p < \infty$, we have*

$$\|f\|_p \leq S\sqrt{p}\|f\|_2.$$

Proof. Let f be an E -polynomial, where E is a Sidon set with constant S . For $x \in \mathbb{T}$, define the random E -polynomial

$$s(x) = \sum_{n \in E} \epsilon_n \hat{f}(n) e^{2\pi i n x},$$

where ϵ_k are independent Rademacher variables. Also, define

$$\sigma = \left(\sum_n |\hat{f}(n)|^2 \right)^{1/2} = \|f\|_2$$

by Parseval's identity. As shown in the implication (iii) $\Rightarrow$ (iv) of the previous theorem, there exists a measure μ on $\mathbb{T}$ such that

$$\|\mu\| \leq S \quad \text{and} \quad \forall n \in E, \quad \hat{\mu}(n) = \epsilon_n.$$

Since $\hat{s} \cdot \hat{\mu} = \hat{f}$, we conclude $f = s * \mu$. By Young's inequality (with $1/p + 1/q = 1 + 1/r$ for $r = p, q = 1$), we get:

$$\|f\|_p \leq \|s\|_p \|\mu\| \leq S\|s\|_p.$$

Assume $p = 2m$, we state the following lemma, which will be proven later:

Lemma 4.1. *For every $x \in \mathbb{T}$, we have*

$$\mathbb{E}(|s(x)|^{2m}) \leq m^m \sigma^{2m}.$$

Raising the Young's inequality to the $2m$ -th power and taking expectation yields:

$$\|f\|_{2m}^{2m} = \mathbb{E}(\|f\|_{2m}^{2m}) \leq S^{2m} \mathbb{E}(\|s\|_{2m}^{2m}) = S^{2m} \int_{\mathbb{T}} \mathbb{E}(|s(x)|^{2m}) dx \leq S^{2m} m^m \sigma^{2m}.$$

Thus,

$$\|f\|_{2m} \leq S\sqrt{m}\|f\|_2.$$

To complete the proof for $2 < p < \infty$, observe that if $p \in [2m - 2, 2m]$, then

$$\|f\|_p \leq \|f\|_{2m} \leq S\sqrt{m}\|f\|_2 \leq S\sqrt{p}\|f\|_2,$$

since $m \leq p$, and the left inequality follows from Hölder's inequality. This proves the result for all $2 < p < \infty$. $\square$

We now prove the lemma.

Proof of Lemma 3.1. Using Chernoff bounds, one can derive a version of the Khintchine inequality:

$$\mathbb{E} [|s(x)|^r] \leq 2^{r+1} r \Gamma\left(\frac{r}{2}\right) \sigma^r,$$

where $r \in (0, \infty)$, Γ denotes the Gamma function, and σ is the standard deviation. For even integers $r = 2m$, Stirling's approximation gives

$$\Gamma(m) = (m-1)! = \frac{m!}{m} \sim \sqrt{2\pi m} m^{m-1} e^{-m},$$

so the bound becomes

$$\mathbb{E} [|s(x)|^{2m}] \lesssim \sqrt{m} \left(\frac{4}{e}\right)^m m^m \sigma^{2m}.$$

Since $\frac{4}{e} > 1$, the previous estimate is not sufficient, and we need to improve the Khintchine bound. It can actually be shown that a stronger version of the Khintchine inequality holds in the case $r > 2$:

$$\mathbb{E} [|s(x)|^r] \leq \frac{2^{r/2}}{\sqrt{\pi}} \Gamma\left(\frac{r+1}{2}\right) \sigma^r,$$

which allows us to replace $\frac{4}{e} > 1$ with $\frac{2}{e} < 1$, and this yields the desired conclusion. However, this result is far from trivial; a complete proof can be found in [4], p. 265.

An alternative argument based on Zygmund's book is also possible and will be explained here. Since s is an E -polynomial, it has only finitely many nonzero coefficients. Therefore, for N large enough, truncating the sum to the interval $[-N, N]$ does not alter s . We denote this truncated sum by s_N . Let us now introduce the notation

$$c_n^f(x) := \left| \widehat{f}(n) e^{2\pi i n x} \right|.$$

Then:

$$\begin{aligned} \mathbb{E} (|s_N(x)|^{2m}) &= \mathbb{E} \left(\left| \sum_{n \leq N} \epsilon_n \widehat{f}(n) e^{2\pi i n x} \right|^{2m} \right) \\ &\leq \mathbb{E} \left(\sum \frac{(\alpha_1 + \dots + \alpha_\ell)!}{\alpha_1! \dots \alpha_\ell!} (c_{n_1}^f(x))^{\alpha_1} \dots (c_{n_\ell}^f(x))^{\alpha_\ell} \epsilon_{l_1}^{\alpha_1} \dots \epsilon_{l_\ell}^{\alpha_\ell} \right), \end{aligned}$$

where the sum is taken over indices $n_1, \dots, n_\ell$ and exponents $\alpha_1, \dots, \alpha_\ell$ satisfying:

- $0 \leq n_i \leq N$,
- $0 \leq \alpha_i \leq 2m$, for $i = 1, \dots, \ell$,
- $1 \leq \ell \leq 2m$,
- $\alpha_1 + \dots + \alpha_\ell = 2m$.

Though the expression may seem complicated, it is just the multinomial expansion.

If any of the α_i are odd, then $\mathbb{E}(\epsilon_{n_1}^{\alpha_1} \cdots \epsilon_{n_\ell}^{\alpha_\ell}) = 0$, due to the symmetry of the Rademacher variables. Therefore, we may assume that each α_i is even, and we write $\alpha_i = 2\beta_i$. Now, let us observe that:

$$\sum \frac{(\beta_1 + \cdots + \beta_\ell)!}{\beta_1! \cdots \beta_\ell!} (c_{n_1}^f(x))^{2\beta_1} \cdots (c_{n_\ell}^f(x))^{2\beta_\ell} = \left(\sum_{|n| \leq N} (c_n^f(x))^2 \right)^m = \sigma^{2m}$$

Now, consider the ratio of the multinomial coefficients:

$$\frac{\frac{(2\beta_1 + \cdots + 2\beta_\ell)!}{(2\beta_1)! \cdots (2\beta_\ell)!}}{\frac{(\beta_1 + \cdots + \beta_\ell)!}{\beta_1! \cdots \beta_\ell!}} = \frac{(2m)!}{m!} \cdot \frac{\beta_1! \cdots \beta_\ell!}{(2\beta_1)! \cdots (2\beta_\ell)!}.$$

Let us focus on the following term:

$$\frac{(2\beta_1)! \cdots (2\beta_\ell)!}{\beta_1! \cdots \beta_\ell!}$$

We claim this term is bounded below by 2^m . Indeed, when $\beta_1 = \cdots = \beta_\ell = 1$ (i.e., $\ell = m$), the ratio is exactly 2^m . Now, assume that for some index i , $\beta_i > 1$, so $\ell < m$. We decompose β_i into two integers $\beta_{i_0} = 1$ and $\beta_{i_1} = \beta_i - 1 \geq 1$. Then:

$$\frac{(2\beta_{i_0})!(2\beta_{i_1})!}{\beta_{i_0}!\beta_{i_1}!} = 2 \frac{(2\beta_i - 2)!}{(\beta_i - 1)!} \leq \frac{(2\beta_i)!}{\beta_i!},$$

which implies that this decomposition decreases the ratio. By iterating this process, we show that the minimal value of the full ratio is 2^m . Finally, observe that:

$$\frac{(2m)!}{m! \cdot 2^m} = \frac{(m+1)(m+2) \cdots (2m)}{2^m} \leq m^m,$$

which completes the argument. □

Remark 4.1. This estimate is sharp: there exists a choice of signs $\pm$ such that the E -polynomial

$$f(x) = \sum_{n=1}^p \pm e^{2\pi i n x}$$

satisfies

$$\|f\|_p \geq \frac{1}{4} \sqrt{p} \|f\|_2,$$

as shown in [1, Section 3.3, p. 212].

5 Application to Exponential Integrability

We first observe that every E -function belongs to $L^2(\mathbb{T})$, and that the L^p -norm inequalities previously established for E -polynomials can be extended to all E -functions. To this end, we first make use of the log-convexity of L^p -norms. Let f be an E -polynomial. For all $p > 2$, there exists $\theta \in (0, 1)$ such that

$$\|f\|_2 \leq \|f\|_1^{1-\theta} \|f\|_p^\theta.$$

Since f is an E -polynomial, the L^p -norm inequality previously proved gives $\|f\|_p \lesssim_p \|f\|_2$. Therefore, combining this with the log-convexity inequality, we obtain $\|f\|_2 \lesssim_p \|f\|_1$. In particular, there exists a constant $C > 0$, such that

$$\|f\|_2 \leq C \|f\|_1.$$

Now let f be a general E -function. In particular, $f \in L^1(\mathbb{T})$, and by Fejér's theorem, the Fejér means $\sigma_n f$ converge to f almost everywhere on $\mathbb{T}$ and in $L^1(\mathbb{T})$. This justifies the application of Fatou's lemma in the following computation:

$$\|f\|_2^2 = \int_{\mathbb{T}} |f|^2 = \int_{\mathbb{T}} \liminf_{n \rightarrow \infty} |\sigma_n f|^2 \leq \liminf_{n \rightarrow \infty} \int_{\mathbb{T}} |\sigma_n f|^2 \leq C^2 \liminf_{n \rightarrow \infty} \|\sigma_n f\|_1^2 = C^2 \|f\|_1^2.$$

Since $f \in L^1(\mathbb{T})$, this shows that every E -function also belongs to $L^2(\mathbb{T})$. A similar argument shows that the L^p -inequalities extend to E -functions:

$$\|f\|_p^p = \int_{\mathbb{T}} |f|^p = \int_{\mathbb{T}} \liminf_{n \rightarrow \infty} |\sigma_n f|^p \leq \liminf_{n \rightarrow \infty} \int_{\mathbb{T}} |\sigma_n f|^p \leq S^p p^{p/2} \liminf_{n \rightarrow \infty} \|\sigma_n f\|_2^p = S^p p^{p/2} \|f\|_2^p.$$

Now, let us use this result to establish exponential integrability.

Corollary 5.1. *Let $E \subset \mathbb{Z}$ be a Sidon set. Then for every E -function f , we have*

$$\int_{\mathbb{T}} \exp(|f(x)|^2) dx < \infty.$$

Proof. Let s_n denote the n -th partial Fourier sum of f . By convexity of $y \mapsto y^2$, we have:

$$|f|^2 \leq 2|f - s_n|^2 + 2|s_n|^2,$$

so that:

$$\exp(|f(x)|^2) \leq \exp(2|f(x) - s_n(x)|^2) \cdot \exp(2|s_n(x)|^2).$$

Fix $\delta > 0$. Since $f \in L^2(\mathbb{T})$, by Parseval's identity, we can choose N such that

$$\|f - s_N\|_2 < \delta.$$

Since s_N is a trigonometric polynomial, it is bounded. Therefore:

$$\int_{\mathbb{T}} \exp(|f(x)|^2) dx \lesssim_N \int_{\mathbb{T}} \exp(2|f(x) - s_N(x)|^2) dx,$$

and we may expand the exponential in power series:

$$\int_{\mathbb{T}} \exp(2|f - s_N|^2) dx = \sum_{p \geq 0} \frac{2^p}{p!} \int_{\mathbb{T}} |f - s_N|^{2p} dx.$$

By the previous theorem, for all $p > 2$:

$$\|f - s_N\|_{2p}^{2p} \leq (S\|f - s_N\|_2)^{2p} p^p \leq (S\delta)^{2p} p^p.$$

So the full series is bounded (up to a multiplicative constant) by:

$$\sum_{p > 2} \frac{(S\delta)^{2p} p^p}{p!}.$$

Using Stirling's formula $p! \sim e^{-p} p^p \sqrt{2\pi p}$, we estimate:

$$\frac{(S\delta)^{2p} p^p}{p!} \sim \frac{(eS^2\delta^2)^p}{\sqrt{p}}.$$

Choosing $\delta > 0$ such that $eS^2\delta^2 < 1$, the series converges and the integral is finite.

□

6 A Norm Characterization of Sidon Sets

In this section, we outline the main ideas behind a probabilistic approach introduced by Gilles Pisier [5], which characterizes Sidon sets through L^p -norm inequalities. We restrict ourselves to the case $G = \mathbb{T}$ and provide only brief sketches rather than detailed proofs. In the following, $E \subset \mathbb{Z}$ denotes a given subset.

Theorem 6.1 (Pisier [5]). *Suppose there exists a constant $S > 0$ such that for every E -polynomial f and all $p \in (2, \infty)$,*

$$\|f\|_p \leq S\sqrt{p}\|f\|_2.$$

Then E is a Sidon set.

Remark 6.1. Thanks to Fatou's lemma, it is sufficient to assume that f is an E -function in $L^2(\mathbb{T})$, not necessarily a trigonometric polynomial. This is the same idea used at the beginning of Section 5.

Before sketching the proof, we present some preliminary results.

Proposition 6.1 (Fernique [7], Preston [8]). *Let $f = \sum_{n \in E} a_n \chi_n \in L^2(\mathbb{T})$ be an E -function, where the χ_n are the characters on the torus $\mathbb{T}$.*

Let $(g_n)_{n \in E}$ be a sequence of independent, centered, normalized Gaussian random variables, assume that E satisfies the above L^p -norm inequality, and that the series $\sum_{n \in E} a_n g_n \chi_n$ converges almost surely to a continuous function on $\mathbb{T}$.

Then, certain exponential integrability properties hold. In particular, due to results of Dudley and Fernique, an entropy condition is satisfied. As a consequence, the original function f is almost everywhere equal to a continuous function on $\mathbb{T}$, which is the key result for our purposes.

Remark 6.2. The proof follows the general strategy outlined in Section 5: one expands the exponential function into its power series, applies the L^p moment inequalities term-by-term, and estimates the resulting series using Stirling's approximation. Then, results of Preston and Fernique ensure the desired almost everywhere continuity.

We now invoke Rider's characterization to connect the almost sure continuity of Gaussian series with the Sidon property.

Proposition 6.2 (Rider [6]). *A set $E \subset \mathbb{Z}$ is a Sidon set if and only if there exists a constant $S > 0$ such that for every finite sequence $(a_n)_{n \in E} \subset \mathbb{C}^E$,*

$$\sum_{n \in E} |a_n| \leq S \mathbb{E} \left\| \sum_{n \in E} a_n g_n \chi_n \right\|_{\infty}.$$

We now turn to sketching the proof of Pisier's theorem.

Sketch of the proof. Let $E \subset \mathbb{Z}$ satisfy the L^p -norm inequality. Define the space

$$C_E^{\text{a.s.}}(\mathbb{T}) := \left\{ f \in L^2(\mathbb{T}) : \sum_{n \in E} \hat{f}(n) g_n \chi_n \text{ converges in } C(\mathbb{T}) \text{ almost surely} \right\}.$$

According to results by Kahane [9], this space can be equipped with the norm

$$[|f|] := \sup_{\substack{A \subset E \\ |A| < \infty}} \mathbb{E} \left\| \sum_{n \in A} \hat{f}(n) g_n \chi_n \right\|_{\infty},$$

making it a Banach space.

Choose $f \in C_E^{\text{a.s.}}(\mathbb{T})$ and denote its Fourier coefficients by $(a_n)_{n \in E}$ for simplicity. By computations detailed in [9, p. 18], the series

$$\sum_{n \in E} |a_n| g_n \chi_n$$

converges almost surely in $C(\mathbb{T})$. Moreover, Parseval's identity implies that $\sum_{n \in E} |a_n|^2 < \infty$, since E satisfies the L^p -norm inequality, we can apply Proposition 1 to the function $\sum_{n \in E} |a_n| \chi_n$. This implies that it coincides almost everywhere, in fact everywhere, with a continuous function on $\mathbb{T}$. Hence, by the uniform convergence of Fejér means, the Fourier series

$$\sum_{n \in E} |a_n| \chi_n$$

converges uniformly, and in particular absolutely. Therefore,

$$\sum_{n \in E} |a_n| < \infty,$$

i.e., $(a_n)_{n \in E} \in \ell^1(E)$. We now define the linear map

$$\begin{aligned} T &: (C_E^{\text{a.s.}}(\mathbb{T}), [|\cdot|]) \longrightarrow (\ell^1(E), \|\cdot\|_1) \\ &f \mapsto (a_n)_{n \in E} \end{aligned}$$

Remark 6.3. The fact that the map T is well defined follows directly from the previous argument, which shows that $(a_n)_{n \in E} \in \ell^1(E)$ for every $f \in C_E^{\text{a.s.}}(\mathbb{T})$.

Both spaces are Banach spaces. By the Closed Graph Theorem, T is continuous if and only if the following implication holds: whenever $f_k \rightarrow f$ in $C_E^{\text{a.s.}}(\mathbb{T})$ and $T(f_k) \rightarrow a$ in $\ell^1(E)$, then $T(f) = a$. This is easily checked. Hence, T is continuous, and there exists a constant $S > 0$ such that

$$\sum_{n \in E} |a_n| \leq S[|f|] = S \mathbb{E} \left\| \sum_{n \in E} a_n g_n \chi_n \right\|_{\infty},$$

which is exactly the characterization of Sidon sets given by Rider. □

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